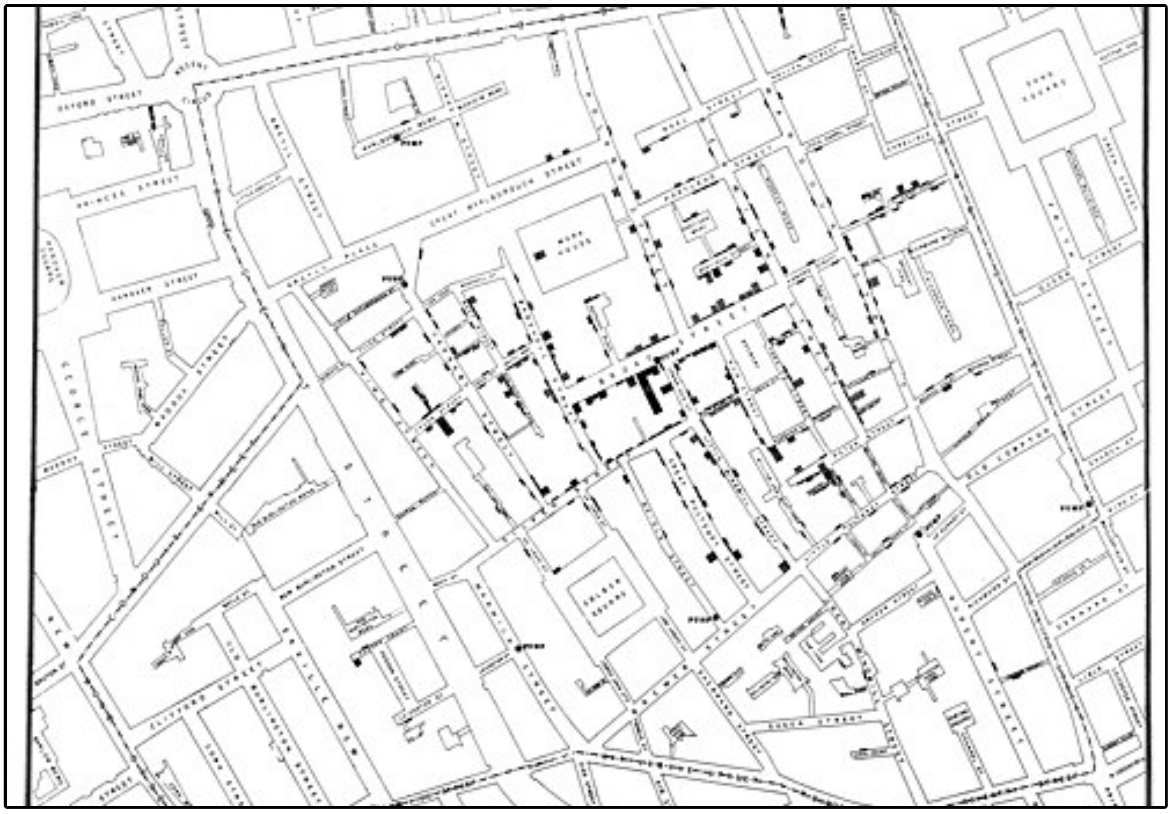
WATER & SANITATION
THROUGH TIME

Why did it take two thousand years to clean up Britain?



Pupil Name: _____ **Teaching Group:** _____

Teacher: _____

Target Grade: _____

Milestone 1 • c.AD 75

Roman Aqueducts & Fishbourne Lead Pipes

Romans construct monumental gravity-fed aqueducts, thermae, and jointed lead pipes at Fishbourne Palace.

Dual-Coding Sketchpad: Draw stone aqueduct arches with water flowing into lead pipes

Milestone 2 • c.1165

Canterbury Monastic Conduits & Settling Tanks

Monk Wibert designs five settling tanks and lead conduits to bring filtered spring water to Christ Church Priory.

Dual-Coding Sketchpad: Draw five settling tanks filtering mud and sand

Milestone 3 • 1596

Sir John Harington: The Flush Toilet

Elizabeth I's godson invents the world's first valve-operated flushing water closet, though lack of running water prevents wide adoption.

Dual-Coding Sketchpad: Draw an overhead cistern with a pull lever flushing into a pan

Milestone 4 • 1831–1842

Asiatic Cholera & Chadwick's 1842 Report

Cholera sweeps through industrial boomtowns; Edwin Chadwick proves squalor causes preventable poverty.

Dual-Coding Sketchpad: Draw back-to-back terraced houses and a shared privy midden

Milestone 5 • 1858–1865

The Great Stink & Bazalgette's Sewers

A boiling heatwave turns the Thames into an unbearable sewer, driving MPs to fund 82 miles of brick intercepting sewers.

Dual-Coding Sketchpad: Draw an egg-shaped brick underground sewer tunnel with flowing water

Milestone 6 • 1854–1884

John Snow, Germ Theory & Robert Koch

Dr John Snow removes the Broad Street pump handle, Pasteur proves Germ Theory, and Koch isolates Vibrio cholerae.

Dual-Coding Sketchpad: Draw a street pump with handle removed and a cholera spot map

Was Roman water engineering built to protect ordinary people, or to show off imperial power and luxury?

1 [1.1] AD 43 · How did the Roman invasion of AD 43 trigger an urban sanitation crisis?	
• Dispersed Celtic Roundhouses	
• Four Imperial Legions	
• Cramped Walled Cities	

2 [1.2] c.AD 43–100 · How did Roman engineers use gravity to transport millions of gallons of water?	
• Distant Mountain Springs	
• Mathematical Downward Gradient	
• Castella Water Towers	

3 [1.3] c.AD 100 · How did monumental public bathhouses promote cleanliness and imperial culture?	
• Subterranean Hypocaust Heating	
• Olive Oil & Bronze Strigils	
• Frigidarium Cold Plunge	

4 [1.4] c.AD 150 · Why did forensic science expose severe flaws in Roman hygiene and health?	
• Communal Sponge on a Stick	
• Intestinal Parasite Eggs	
• Toxic Lead Water Pipes	

5 [1.5] AD 410 · Why did Roman water engineering abruptly collapse after AD 410?	
• Imperial Power & Vanity	
• Vanishing Roman Legions	
• Thousand-Year Sanitation Void	

Roman lead pipes at Fishbourne carried spring water under gravity pressure to garden fountains.

Question 2: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why the Romans built sophisticated water supply and sanitation systems in Britain after AD 43. [12 marks]

Stimulus: **The need for fresh water and hygiene in densely populated towns like Londinium** | **Displaying imperial power, Romanitas, and luxury at sites like Fishbourne Palace** *(You must also use information of your own)*

1. URBAN DENSITY & HYGIENE	2. IMPERIAL POWER & ROMANITAS	3. MILITARY READINESS & VERDICT
Explain how founding crowded stone towns (Londinium, York) forced Romans to construct gravity aqueducts and flushing latrines to prevent devastating epidemics [1.1, 1.2].	Explain how grand thermae, heated hypocausts, and luxury lead pipes at Fishbourne Palace were designed to display imperial supremacy (*romanitas*) [1.3, 1.5].	Weigh whether practical military survival or imperial vanity was more decisive in driving Roman engineering [1.4, 1.5].

KEY TERMS: **Aqueduct • Thermae • Hypocaust • Strigil • Latrine • Tersorium • Plumbism • Romanitas**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 1). Complete the dual-coding sketch challenge in the timeline sketchpad.

Was medieval public health squalor caused by a lack of money, or by mistaken medical beliefs like miasma?

1 [2.1] 1100–1250 · Why did the growth of medieval towns overwhelm traditional water supplies?	
• Collapsed Roman Stone Aqueducts	
• Booming Market Boroughs	
• Unpaved Muddy Thoroughfares	

2 [2.2] c.1250–1348 · How did medieval towns attempt to manage human and animal waste?	
• Overcrowded Timber Tenements	
• Night-Time Gong Farmers	
• Court Leet Fines & Rakers	

3 [2.3] c.1300 · Why did medieval monasteries enjoy far cleaner water than crowded towns?	
• Settling Tanks & Lead Conduits	
• Segregated Monastic Cloisters	
• Titchfield Abbey Fishponds	

4 [2.4] 1348–1350 · How did the 1348 Black Death challenge medieval understanding of disease?	
• Yersinia Pestis in Fleas	
• Poisonous Foul Air (Miasma)	
• Forty-Day Port Quarantines	

5 [2.5] 1388 · Did public health genuinely collapse, or have Victorians exaggerated medieval squalor?	
• Statute of Cambridge (1388)	
• Compulsory River Clean-Up	
• Carole Rawcliffe Revisionism	

Medieval monks drank spring water filtered through five settling tanks while townspeople drank from rivers.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why living conditions and public health in medieval towns were so much worse than in medieval monasteries. [12 marks]

Stimulus: **The design and funding of monastic water systems such as Canterbury Priory** | **The rapid, unregulated growth of commercial towns and lack of central taxation** *(You must also use information of your own)*

1. TOWN SQUALOR & LACK OF INFRASTRUCTURE	2. MONASTIC WEALTH & HYDRAULIC PLANNING	3. COMPARATIVE VERDICT
Explain how rapid population growth, unpaved streets, animal butchery, and leaking cesspits contaminated town water supplies [2.1, 2.2].	Explain how vast land endowments enabled monasteries (Canterbury, Titchfield) to construct settling tanks, lead conduits, and flushing reredorters [2.3].	Weigh whether financial wealth or geographical isolation was more decisive in explaining the health divide between towns and monasteries [2.3, 2.5].

KEY TERMS: **Borough • Gong Farmer • Cesspit • Miasma • Settling Tank • Lavatorium • Black Death • Statute of Cambridge**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 2). Complete the dual-coding sketch challenge in the timeline sketchpad.

Was early modern filth caused by population explosion, or the lack of running water to make inventions work?

1 [3.1] 1500–1600 · How did London's rapid expansion create an unprecedented sanitation crisis?	
• National Population Doubles	
• 500,000 London Residents	
• Cantilevered Timber Tenements	

2 [3.2] 1596 · Why did Sir John Harington's revolutionary flush toilet fail to clean up Britain?	
• Queen Elizabeth's Godson	
• Valve-Operated Cistern	
• Absence of Running Water Mains	

3 [3.3] 1609–1613 · How did private enterprise attempt to bring clean spring water into London?	
• Sir Hugh Myddelton	
• 38-Mile Man-Made Canal	
• Hollowed Elm Tree Trunk Pipes	

4 [3.4] 1660–1665 · What does Samuel Pepys' diary reveal about domestic sanitation in Stuart London?	
• Stepping into Neighbour's Waste	
• Fractured Brick Cesspits	
• Nightsoil Men & Filth Carts	

5 [3.5] 1665 · Why did medical responses to the 1665 Great Plague remain fundamentally medieval?	
• 100,000 Plague Fatalities	
• Red Crosses & Watchmen	
• Pounding Tobacco & Pomanders	

Sir John Harington invented the flush toilet in 1596, but towns lacked running water mains to make it work.

Question 3: Evaluative Statement Essay [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

'Early Modern towns made substantial progress in water supply and sanitation between 1500 and 1700.' How far do you agree? [12 marks]

Stimulus: **Inventions and infrastructure like Harington's water closet (1596) and Myddelton's New River (1613)** | **Domestic conditions documented by Samuel Pepys and the devastating Great Plague of 1665** | *(You must also use information of your own)*

1. TECHNOLOGICAL PROGRESS & INNOVATION	2. POPULATION PRESSURE & SQUALOR	3. SUSTAINED HISTORICAL VERDICT
Explain how Harington's flush toilet (1596) and Myddelton's New River (1613) demonstrated genuine engineering progress [3.2, 3.3].	Explain how explosive population growth, overflowing cesspits (Pepys), and the 1665 Great Plague proved ordinary conditions worsened [3.1, 3.4, 3.5].	Weigh whether private elite breakthroughs or widespread public squalor defined the era [3.5].

KEY TERMS: **Metropolis • Water Closet • Sir John Harington • New River • Samuel Pepys • Searchers of the Dead • Nightsoil • Bills of Mortality**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 3). Complete the dual-coding sketch challenge in the timeline sketchpad.

Samuel Pepys recorded stepping into his cellar in October 1660 to find his neighbour's cesspit had leaked through.

Was Victorian cholera spread by greedy factory owners building cheap slums, or by the government's refusal to interfere?

1 [4.1] 1780–1830 · How did industrial urbanisation create catastrophic living conditions?	
• Steam-Powered Mill Towns	
• Back-to-Back Slum Rows	
• 50 People Sharing One Privy	

2 [4.2] c.1830 · Why was drinking water in industrial towns dangerously contaminated?	
• Shallow Garden Wells	
• Leaking Privy Cesspools	
• Intermittent Standpipes	

3 [4.3] 1831–1832 · Why did the arrival of cholera in 1831 cause widespread national terror?	
• Sunderland Coal Port Arrival	
• Dehydration in Hours	
• Miasma Air Purification Fires	

4 [4.4] 1842 · How did Edwin Chadwick prove that slum filth was costing Britain a fortune?	
• 400-Page Parliamentary Audit	
• Working-Class Life Expectancy 17	
• Filth Causes Preventable Poverty	

5 [4.5] 1848–1854 · Why did Victorian ideology fiercely resist compulsory government intervention?	
• Laissez-Faire ('Leave Alone')	
• Ratepayer Taxpayer Protests	
• Sacking Chadwick in 1854	

Edwin Chadwick proved in 1842 that industrial labourers in Manchester died at an average age of just 17.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why public health in industrial towns deteriorated into a national crisis in the period 1800–1850. [12 marks]

Stimulus: **The rapid rate of urbanisation and construction of back-to-back housing** | **The Victorian political belief in laissez-faire and opposition from ratepayers** *(You must also use information of your own)*

1. UNCONTROLLED URBANISATION & HOUSING	2. WATER CONTAMINATION & CHOLERA	3. LAISSEZ-FAIRE & POLITICAL GRIDLOCK
Explain how rapid population migration created back-to-back housing, damp cellar dwellings, and overflowing privy middens without sewers [4.1, 4.2].	Explain how shallow wells contaminated with sewage collided with the 1831 arrival of Asiatic Cholera and miasma dogma [4.2, 4.3].	Explain how the ideology of laissez-faire and ratepayer resistance crippled the 1848 Public Health Act [4.4, 4.5].

KEY TERMS: **Urbanisation • Back-to-back housing • Privy Midden • Asiatic Cholera • Edwin Chadwick • Public Health Act 1848 • Laissez-Faire • Ratepayer**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 4). Complete the dual-coding sketch challenge in the timeline sketchpad.

Did politicians clean up the Thames out of care for poor people, or out of pure panic for their own lives?

1 [5.1] 1850–1855 · How did the invention of the water closet accidentally destroy the River Thames?

• Flushing Toilets Direct to River	
• Michael Faraday's Opaque River Test	
• Fermenting Low-Tide Mudflats	

2 [5.2] June 1858 · How did the Great Stink of 1858 force Parliament to act in sheer panic?

• 35°C Blistering Summer Heat	
• Curtains Soaked in Lime Chloride	
• MPs Fleeing River Committee Rooms	

3 [5.3] 1859–1865 · How did Joseph Bazalgette engineer the greatest sanitation system in the world?

• 318 Million Portland Cement Bricks	
• Oval Egg-Shaped Self-Flushing Sewers	
• Crossness & Abbey Mills Pumping	

4 [5.4] 1861–1865 · How did Louis Pasteur's 1861 Germ Theory overturn centuries of miasma dogma?

• Boiled Broth in Curved Glass Flasks	
• Airborne Microbes Disprove Spontaneous Generation	
• Eastney Pumping Station (1865)	

5 [5.5] 1875 · Why was the 1875 Public Health Act the decisive turning point in British sanitation?

• Compulsory Medical Officers of Health	
• Mandatory Street Lighting & Sewers	
• Final Death of Laissez-Faire	

Michael Faraday warned in 1855 that the Thames was an opaque, fermenting sewer of human waste.

Question 2: Explain Why [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why the British government finally took decisive action to build London's sewer network after 1858. [12 marks]

Stimulus: **The Great Stink of June 1858 and its impact on politicians in Parliament** | **The engineering proposals of Joseph Bazalgette and fear of recurring cholera epidemics** *(You must also use information of your own)*

1. THE GREAT STINK & PARLIAMENTARY PANIC
Explain how the blistering June 1858 heatwave cooked Thames sewage, driving MPs from committee rooms and forcing an emergency sewer bill in 18 days [5.1, 5.2].

2. BAZALGETTE'S BLUEPRINT & CHOLERA FEAR
Explain how Joseph Bazalgette's ready-to-build plans and recurring cholera epidemics gave politicians a proven engineering solution [5.3].

3. COMPARATIVE VERDICT
Weigh whether selfish fear of dying or long-term sanitary vision was more decisive in breaking the laissez-faire gridlock [5.2, 5.5].

KEY TERMS: **The Great Stink • Joseph Bazalgette • Intercepting Sewer • Portland Cement • Beam Engine • Louis Pasteur • Germ Theory • Public Health Act 1875**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 5). Complete the dual-coding sketch challenge in the timeline sketchpad.

Did John Snow end cholera through medical authority, or through meticulous geographical detective work?

1 [6.1] August 1854 · How did the sudden cholera outbreak of August 1854 plunge Soho into catastrophe?

- **Broad Street Intersection**
- **500 Fatalities in Ten Days**
- **Medical Establishment Miasma Dogma**

2 [6.2] September 1854 · Why did Dr John Snow suspect water rather than foul air?

- **Royal Anesthetist to Queen Victoria**
- **Gut-First Symptoms Rather than Lungs**
- **Drinking Supply Theory**

3 [6.3] September 1854 · How did John Snow use scientific detective work and mapping to prove his case?

- **616 Cholera Death Bars Plotted**
- **Brewery Workers Drinking Malt Beer**
- **Removing the Broad Street Pump Handle**

4 [6.4] 1854–1855 · What microscopic contamination was finally discovered beneath the pump?

- **Baby Lewis's Soiled Diaper**
- **Cracked Brickwork in 40 Broad Street**
- **Cesspool Sewage Leaking into Well**

5 [6.5] 1855–1884 · Why did medical authorities resist John Snow's findings for decades?

- **General Board of Health Rejection**
- **Robert Koch Isolates Vibrio Cholerae (1884)**
- **Clean Water as a Human Right**

Dr John Snow proved cholera was water-borne by mapping 616 deaths around the Broad Street pump in Soho.

Question 5: Disciplinary Evaluation [12 marks • 15 mins]

EXTENDED WRITING PRACTICE

Explain why Dr John Snow's investigation of the 1854 Broad Street cholera epidemic was a decisive turning point in medicine. [12 marks]

Stimulus: **The cartographic and geographical evidence of the 1854 Cholera Spot Map** | **The persistence of miasma theory and delayed acceptance by the Board of Health** *(You must also use information of your own)*

1. SCIENTIFIC METHODOLOGY & MAPPING
Explain how Snow used clinical logic, the 1854 Spot Map, the brewery anomaly, and removing the pump handle to pioneer modern epidemiology [6.2, 6.3].

2. DISPROVING MIASMA & REVEALING SEWAGE
Explain how uncovering Baby Lewis's diaper and the leaking cesspool at No. 40 proved that water-borne bacteria, not bad air, caused cholera [6.4].

3. COMPARATIVE VERDICT & LONG-TERM IMPACT
Weigh the immediate rejection by the Board of Health against the ultimate vindication by Robert Koch in 1884 [6.5].

KEY TERMS: **Epidemiology • Dr John Snow • Broad Street Pump • Spot Map • Brewery Anomaly • Baby Lewis • Robert Koch • Point Source Epidemic**

ANALYTICAL CONNECTIVES: *The primary catalyst explaining why... was... • Specifically, this meant that... • In particular, this was demonstrated when... • On the other hand, a decisive underlying factor was... • Consequently, this reveals that... • Ultimately, the decisive reason was... because...*

TIMELINE MISSION Turn to Pages 2–3 (Milestone 6). Complete the dual-coding sketch challenge in the timeline sketchpad.

Robert Koch isolated the Vibrio cholerae bacterium under the microscope in 1884, confirming Snow's theory.

PUPIL NAME: _____ **CLASS:** _____ **TARGET LEVEL:** _____

KS3 PATHWAY	**Emerging (1–2):** Isolated facts; basic descriptive recall.	**Developing (3):** Identifies causes & features with simple explanation.	**Secure (4–5):** Structured PEEL arguments with precise facts.	**Advanced (6–7):** Evaluates competing causes; nuanced explanation.	**Mastery (8–9):** Sustained historical criteria & historiographical verdicts.
EFFORT RUBRIC	**1 • Concern:** Incomplete work.	**2 • Inconsistent:** Needs prompts.	**3 • Satisfactory:** Meets baseline.	**4 • Good:** Thoughtful scholar.	**5 • Exemplary:** Exceptional pride.

#	ENQUIRY TITLE	HISTORICAL SKILL	DO NOW	TASK 4 GRADE	EFFORT	TEACHER FEEDBACK
UNIT SUMMATIVE OUTCOME			**[/0]**	**[ ]**	**[12345]**	**Signed:** _____ • **Date:** ___/___/2026

OVERALL UNIT FORMATIVE ASSESSMENT & ACADEMIC GUIDANCE — KEY STAGE 3 MASTERY

WHAT WENT WELL (WWW):

EVEN BETTER IF (EBI):

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT SELF-MARKING QUIZZES (0 LESSONS • 0 QUESTIONS) — Instant Recall Practice

SUMMER ASSESSMENT REVISION PROTOCOL • 4 EVIDENCE-BASED RETRIEVAL TECHNIQUES — COGNITIVE SCIENCE IN PRACTICE

1. Spaced Flash Quizzing
Scan each QR code weekly. Retest until you score 100% on factual recall before checking your workbook notes.

2. Dual-Coding Timeline Walk
Turn to Pages 2–3. Cover the written text and narrate the historical story aloud using only your sketchpad symbols.

3. 5-Minute Brain Dumps
Pick an enquiry question. Spend 5 uninterrupted minutes writing every name, date, and cause from memory.

4. Causal Connective Drills
Draft 3 PEEL sentences explaining *why* an event happened using: *Consequently... This directly resulted in...*